Project Initialization and Planning Phase

Date	15 March 2024
Team ID	-
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to predict the Human Development Index (HDI) of a country using Machine Learning techniques. The system analyzes important development indicators such as Life Expectancy at Birth, Expected Years of Education, Mean Years of Education, and Gross National Income (GNI) per Capita to estimate the HDI score. This solution provides quick and reliable predictions that can support educational, research, and policy-making purposes.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning-based web application that predicts the Human Development Index (HDI) using key socio-economic indicators. The system provides accurate predictions through a simple and user-friendly interface.
Scope	The project focuses on collecting HDI-related data, preprocessing the dataset, training a Linear Regression model, and deploying the trained model using the Flask framework. Users can enter development indicators through the web application and receive the predicted HDI score along with its development category.
Problem Statement	
Description	Calculating and analyzing the Human Development Index manually requires processing multiple socio-economic indicators. This process is time-consuming and may lead to errors. There is a need for an automated system that can accurately predict HDI based on key development factors.
Impact	The proposed system enables faster and more accurate HDI prediction. It helps students, researchers, and policymakers understand how different socio-economic factors influence human development, supporting better planning and decision-making.
Proposed Solution	

Approach	The system uses a Machine Learning Linear Regression model trained on Human Development Index data. The trained model is integrated into a Flask web application where users can enter development indicators and instantly obtain the predicted HDI score.
Key Features	• Machine Learning-based HDI prediction. • User-friendly web interface developed using Flask. • Fast prediction of Human Development Index. • Displays predicted HDI score and development category. • Uses Life Expectancy, Expected Years of Education, Mean Years of Education, and GNI per Capita as input features. • Simple, accurate, and easy to use.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5 or above
	Memory	8 GB RAM
	Storage	500 GB SSD or above
Software	Framework	Flask
	Programming Language	Python
	Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
	Development Environment	Jupyter Notebook, Visual Studio Code
Data	Source, Size, Format	Human Development Index (HDI) Dataset, CSV Format